Data Analytics Capstone

Gallstone Disease Risk Prediction Using Machine Learning

Final Report

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QM640: Data Analytics Capstone

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GitHub repository

All raw data, processed data, code notebooks, and figures supporting this report are available in the public GitHub repository at:

Repository Link: https://github.com/prasun031087-afk/Capstone-Gallstone-Prediction

Raw data: data/dataset-uci.xlsx

Processed / clean data: data/dataset-uci.xlsx

Code / notebooks: notebooks/QM640_Data_Analytics_Capstone.ipynb

Reports: reports/

QM640 Final Synopsis Prasun Das.pdf

QM640 Interim Report Prasun Das.pdf

QM640 Final Report Prasun Das.pdf

Abstract

Gallstone disease affects more than 700 million people worldwide, but current clinical workflows still detect the condition only after symptoms have appeared, when imaging is ordered. There is no widely deployed, low-cost screening tool that combines routinely collected demographic, biochemical, and bioimpedance information to flag patients who would benefit from earlier intervention. This capstone addresses that gap by building and evaluating reproducible supervised classification models that estimate gallstone risk at the individual patient level and by surfacing the most influential risk drivers through formal hypothesis testing.

The study uses the publicly available UCI Gallstone-1 dataset (Esen et al., 2024), which captures cross-sectional records for 319 patients (158 cases and 161 controls) collected at Ankara VM Medical Park Hospital. Each record contains 38 predictors spanning demographics, comorbidities, anthropometry, bioimpedance compartments, body composition, hepatic imaging grade, and serum biochemistry, along with a binary gallstone outcome. The unit of analysis is the individual patient. Analyses follow a four-question design that combines exploratory data analysis, Mann–Whitney U and chi-square hypothesis tests with Benjamini–Hochberg false discovery rate correction, Spearman correlation across WHO body mass index categories, and supervised machine learning.

All work is implemented in Python on a standard CPU environment using pandas, NumPy, SciPy, scikit-learn, and matplotlib/seaborn. Two complementary classifiers are compared: a standardized logistic regression baseline and a 400-tree balanced random forest. Models are evaluated with five-fold stratified cross-validation on 80 percent of the data and a single held-out 20 percent test set.

The trained random forest reaches a cross-validated area under the curve of 0.839 (logistic regression 0.825) and a held-out test AUC of 0.867 (logistic regression 0.881), exceeding the 0.84 benchmark reported by Esen et al. (2024). Feature importance and standardized coefficients converge on C-reactive protein, vitamin D, lean mass percentage, total body fat ratio, hepatic fat accumulation, and hyperlipidemia as the dominant signals. Obesity defined solely by body mass index is not significantly associated with gallstones, indicating that detailed body composition adds clinically meaningful information beyond a simple weight-for-height threshold.

The pipeline is suitable for embedding electronic health record screening dashboards, primary-care risk calculators, and outpatient triage workflows, giving clinicians a transparent, low-cost decision support layer that can identify high-risk patients before symptomatic presentation.

Introduction

Gallstone disease, also known as cholelithiasis, is a common digestive disorder caused by imbalances in bile components such as cholesterol or bilirubin, leading to hardened deposits in the gallbladder. While many of these deposits remain symptomless, they can later cause severe pain, nausea, jaundice, and emergency cholecystectomy. The condition affects more than 700 million people worldwide and continues to grow in prevalence, particularly in urban populations exposed to poor diet, sedentary behavior, and obesity. Because diagnosis still relies on imaging that is typically ordered only after symptoms appear, early identification of high-risk patients remains an open clinical problem and a meaningful opportunity for data-driven decision support.

Background and Context

Gallstones form when bile becomes oversaturated with cholesterol or bilirubin, causing crystal formation in the gallbladder. Major risk factors include age, female gender, obesity, diabetes, hyperlipidemia, rapid weight loss, family history, and unfavorable body composition changes such as higher visceral fat and lower lean mass. Gallstone disease creates a significant healthcare burden because cholecystectomy remains one of the most common abdominal surgeries and costs continue rising with increasing metabolic syndrome prevalence (Lammert et al., 2016; Stinton & Shaffer, 2012).

Routine biochemistry tests, bioelectrical impedance measurements, demographic information, and comorbidity data from electronic health records provide low-cost indicators that can support supervised machine learning models for early gallstone risk prediction in primary-care and outpatient settings. Key stakeholders include physicians, hospital screening programs, insurers, and patients who may benefit from earlier intervention.

Although research shows that classical machine learning models can effectively predict gallstone risk (Esen et al., 2024), there is still no widely adopted clinical decision support system that integrates demographic, biochemical, and bioimpedance data into a unified gallstone risk assessment tool.

Problem Statement

The objective of this study is to predict Y = gallstone status (1 = present, 0 = absent) for individual adult patients using X = a structured set of 38 routinely collected predictors that span demographics, comorbidities, anthropometry, bioimpedance compartments, body composition, hepatic fat imaging grade, and serum biochemistry, captured cross-sectionally at Ankara VM Medical Park Hospital and released through the UCI Machine Learning Repository in 2024. Success is evaluated using the area under the receiver operating characteristic curve (AUC), accuracy, precision, recall, and F1 score, with the AUC obtained from the Esen et al. (2024) publication serving as the external benchmark.

Purpose of the study

The purpose of this study is to use the publicly available UCI Gallstone-1 dataset to build and evaluate proof-of-concept supervised classification models that predict gallstone status from demographic, comorbidity, body composition, and serum biochemistry features. In parallel, the study aims to identify the strongest individual risk drivers through formal hypothesis testing, characterise how body composition varies across body mass index categories, and quantify whether obesity alone explains gallstone presence. The deliverable is a reproducible analytical pipeline that may inform the future design of clinical decision support systems and screening protocols rather than a tool intended for direct patient use.

Research Questions

The four research questions along with the hypothesis that guide the study are stated below.

Research Question 1 (RQ1): What patterns exist in the distribution of clinical, biochemical, and bioimpedance features of the Gallstone dataset, and how do key variables differ between patients with gallstones and patients without gallstones?

Research Question 2 (RQ2): How do body composition metrics such as total body fat ratio, visceral fat rating, lean mass percentage, and muscle mass vary across body mass index categories ranging from underweight to obese, and what observable trends emerge as body mass index increases?

Research Question 3 (RQ3): Is there a statistically significant association between obesity status, defined as a body mass index of thirty kilograms per square meter or higher, and the presence of gallstones in this patient population?

Research Question 4 (RQ4): Can simple machine learning classification models accurately predict whether a patient has gallstones based on their clinical, biochemical, and bioimpedance features, and which features are the most important drivers of this prediction?

Contributions and Expected Value

The study contributes to practical, technical, and stakeholder-focused value by developing a reproducible machine learning framework for early gallstone risk prediction using routinely available clinical and body composition data.

Practical contribution: The study delivers a transparent, low-cost gallstone risk screening pipeline that combines features already collected during routine outpatient visits, enabling clinicians to flag high-risk patients before symptomatic presentation and to triage them for confirmatory imaging.

Technical contribution: The work re-validates the Esen et al. (2024) benchmark using a fully reproducible scikit-learn pipeline that pairs a standardized logistic regression with a balanced random forest, layers in Benjamini–Hochberg adjusted hypothesis testing for 31 continuous and 7 categorical predictors and quantifies feature importance via both Gini and standardized coefficients.

Value to stakeholders: Primary-care physicians and hospital screening programs gain a calibrated probability of gallstone presence and a short list of clinically actionable drivers, payers gain a tool to support preventive care strategies, and patients potentially benefit from earlier lifestyle and pharmacological intervention.

Literature Survey

Eleven sources, six clinical or methodological and five software or statistical, inform the design of this study. Each source contributes to the conceptual, methodological, or analytical framework of the capstone.

Literature review approach

Sources for this study were identified through searches of PubMed, Google Scholar, and the UCI Machine Learning Repository using keyword combinations such as “gallstone disease,” “cholelithiasis,” “risk factors,” “bioimpedance,” “machine learning classification,” “logistic regression,” “random forest,” “SMOTE,” and “false discovery rate.” Inclusion criteria favoured peer-reviewed clinical reviews on gallstone epidemiology, foundational methodological papers on the algorithms and statistical procedures used in the analysis, and computational toolkits required to reproduce the work.

Summary of key literature

Gallstone disease research is grounded in established epidemiological and clinical frameworks. Lammert, F. et al. (2016) provide a widely cited consensus on gallstones. They describe epidemiology, pathophysiology, and the four F risk factors. This supports the importance of early and non-invasive risk stratification. Stinton, L. M. and Shaffer, E. A. (2012) highlight obesity, diabetes, rapid weight loss, and lipid abnormalities as major modifiable risks. These findings inform the feature selection and research questions in this study.

The dataset used originates from Esen, B. et al. (2024). They applied logistic regression, k nearest neighbors, and support vector machines to a cohort of 319 patients. Their reported AUC of about 0.84 provides a useful benchmark for evaluating model performance in this study.

Methodologically, Chawla, N. V. et al. (2002) introduce the SMOTE technique. This method is demonstrated to address potential class imbalance in future datasets. Breiman, L. (2001) establishes the foundation for Random Forest. This supports its selection because it can model non-linear relationships and provide interpretable feature importance. Cohen, J. (1988) informs the study sample size and effect size calculations using established conventions such as Cohen w equals 0.30 and f squared equals 0.15.

From a computational perspective, Pedregosa, F. et al. (2011) provide the basis for the scikit-learn implementation used for preprocessing, model building, and evaluation. McKinney, W. (2010) supports the use of pandas for data manipulation and exploratory analysis. Statistical testing relies on Virtanen, P. et al. (2020) through SciPy. This enables non-parametric and contingency based analyses.

Public health standards are guided by the World Health Organization (2021). This source defines BMI classifications used in the study. Finally, multiple testing correction is handled using the false discovery rate procedure introduced by Benjamini, Y. and Hochberg, Y. (1995). This ensures robust statistical inference.

Table 1

Literature relevance matrix

Author (Year)	**Domain / Context**	**Dataset / Setting**	**Method(s)**	**Key Findings**	**Supports RQ / Decisions**
Lammert et al. (2016)	Clinical review of gallstones	Global epidemiology	Narrative consensus review	Defines four-F risk factors and	Frames RQ1 and motivates risk

Author (Year)	Domain / Context	Dataset / Setting	Method(s)	Key Findings	Supports RQ / Decisions
				epidemiology of cholelithiasis	stratification (RQ4)
Stinton & Shaffer (2012)	Epidemiology of gallbladder disease	Population-level evidence	Systematic review	Obesity, diabetes, lipid abnormalities are major modifiable risks	Justifies feature set for RQ2, RQ3
Esen et al. (2024)	Gallstone risk prediction	319 patients, Ankara cohort	LR, k-NN, SVM	Reported AUC ≈ 0.84 with bioimpedance + biochemistry features	Benchmark for RQ4; provides the dataset
Breiman (2001)	Statistical learning	Methodological	Random Forest theory	Ensemble of trees with feature importance	Model choice for RQ4
Chawla et al. (2002)	Imbalanced learning	Methodological	SMOTE oversampling	Synthetic minority oversampling improves rare-class recall	Mitigation strategy for class imbalance in RQ4
Cohen (1988)	Statistical power analysis	Methodological	Effect-size frameworks	Tabled values for w, f^2	Underpins sample-size calculations
Pedregosa et al. (2011)	Machine learning toolkit	Software	scikit-learn library	End-to-end ML pipelines in Python	Implementation engine for RQ4
McKinney (2010)	Data analysis toolkit	Software	pandas library	Tabular data manipulation in Python	Used throughout RQ1, RQ2 EDA

Author (Year)	Domain / Context	Dataset / Setting	Method(s)	Key Findings	Supports RQ / Decisions
Virtanen et al. (2020)	Scientific computing	Software	SciPy library	Statistical tests, distributions, optimisation	Hypothesis tests for RQ1, RQ3
Benjamini & Hochberg (1995)	Multiple testing	Methodological	False discovery rate	Controls expected proportion of false positives	Adjusts the 38 simultaneous tests in RQ1
WHO (2021)	Public health standards	Global guidelines	BMI classification	Underweight / normal / overweight / obese cutoffs	Defines BMI categories for RQ2, RQ3

Materials and Method

Data Description

The dataset used in this study is the Gallstone dataset published on the University of California Irvine Machine Learning Repository. It is publicly accessible via the UCI repository at https://archive.ics.uci.edu/dataset/1150/gallstone-1. The dataset is provided as a single comma separated values file and can be loaded directly into Python using the pandas library (McKinney, 2010). It is also fully compatible with the scikit learn machine learning library (Pedregosa et al., 2011), which will be used to build and evaluate all classification models in this study. The dataset was collected at Ankara VM Medical Park Hospital and originally released by Esen et al. (2024). Each patient record carries a binary label that indicates whether gallstones were detected during the imaging examination.

Dataset Overview

Number of records (rows): 319 patient records

Number of variables (columns): 39 (38 predictors + 1 binary target)

Time period: Cross-sectional cohort collected at Ankara VM Medical Park Hospital and released through UCI in 2024

Unit of analysis: Individual patient

Target variable: Gallstone Status: binary (1 = confirmed gallstone case, 0 = control patient without gallstones).

Inclusion criteria: All adult patients in the original Ankara cohort with complete demographic, comorbidity, anthropometric, bioimpedance, hepatic imaging, and serum biochemistry measurements were included.

Exclusion criteria: No additional records were excluded after the data integrity audit (no duplicates, no implausible values, and no missing entries were detected).

A complete data dictionary that lists every variable, its data type, allowed values, and grouping is provided in Table 2 below; detailed test outputs and code references are reproduced in Appendix A.

Table 2

Data dictionary in the Gallstone-1 dataset, grouped by role.

Variable Name	Definition	Datatype	Allowed Values/Range	Missing Values Handling	Notes
Gallstone Status	Target varia-ble, Gallstones present (0), and absent (1)	Binary	0 = absent, 1 = present	None observed	Target variable

Age	Age of the person	Integer	Adult range (years, integer)	None observed	Demographics
Gender	Gender of the person	Categorical	0 = Male, 1 = Female	None observed	Demographics
Comorbidity	Concomitant diseases	Categorical	Integer count (0 to several)	None observed	Comorbidity
Coronary Artery Disease	Cardiovascular disease	Binary	0 / 1	None observed	Comorbidity
Hypothyroidism	Underactive thyroid gland	Binary	0 / 1	None observed	Comorbidity
Hyperlipidemia	High levels of fat in the blood	Binary	0 / 1	None observed	Comorbidity
Diabetes Mellitus (DM)	High blood sugar	Binary	0 / 1	None observed	Comorbidity
Height	Height is the length	Integer	Adult range (cm, integer)	None observed	Anthropometry
Weight	Body weight	Continuous	Adult range (kg)	None observed	Anthropometry
Body Mass Index (BMI)	Weight for height ratio	Continuous	Clinical range (kg/m^2)	None observed	Anthropometry
Total Body Water (TBW)	Total water in the body	Continuous	Bioimpedance range (L)	None observed	Bioimpedance
Extracellular Water (ECW)	It is extracellular water	Continuous	Bioimpedance range (L)	None observed	Bioimpedance
Intracellular Water (ICW)	It is intracellular water	Continuous	Bioimpedance range (L)	None observed	Bioimpedance
Extracellular Fluid/Total Body Water (ECF/TBW)	Extracellular water content	Continuous	0–1 (ratio)	None observed	Bioimpedance
Total Body Fat Ratio (TBFR)	Total fat content (units: %)	Continuous	0–100 %	None observed	Bioimpedance
Lean Mass (LM)	Lean body mass (units: %)	Continuous	0–100 %	None observed	Bioimpedance
Body Protein Content (Protein)	Essential building blocks for the body (units: %)	Continuous	0–100 %	None observed	Bioimpedance
Visceral Fat Rating (VFR)	Visceral organ fat level	Integer	Integer (typical 1–30)	None observed	Bioimpedance

Bone Mass-Bone Mass (BM)	Bone weight	Continuous	Bioimpedance range (kg)	None observed	Bioimpedance
Muscle Mass (MM)	Amount of muscle	Continuous	Bioimpedance range (kg)	None observed	Bioimpedance
Obesity	Excessive adiposity (units: %)	Continuous	0–100 %	None observed	Bioimpedance
Total Fat Content (TFC)	Total fat amount	Continuous	Bioimpedance range (kg)	None observed	Bioimpedance
Visceral Fat Area (VFA)	Inner adipose tissue area	Continuous	Bioimpedance range (cm^2)	None observed	Bioimpedance
Visceral Muscle Area (VMA)	Inner muscle area (units: kg)	Continuous	Bioimpedance range (kg)	None observed	Bioimpedance
Hepatic Fat Accumulation (HFA)	Accumulation of fat in the liver	Categorical	Ordinal grade (0–4)	None observed	Imaging
Glucose	Blood sugar	Continuous	Clinical reference range (mg/dL)	None observed	Serum biochemistry
Total Cholesterol (TC)	It is total cholesterol	Continuous	Clinical reference range (mg/dL)	None observed	Serum biochemistry
Low Density Lipoprotein (LDL)	Bad cholesterol	Continuous	Clinical reference range (mg/dL)	None observed	Serum biochemistry
High Density Lipoprotein (HDL)	Good cholesterol	Continuous	Clinical reference range (mg/dL)	None observed	Serum biochemistry
Triglyceride	Type of fat found in the blood	Continuous	Clinical reference range (mg/dL)	None observed	Serum biochemistry
Aspartat Aminotransferaz (AST)	Type of liver enzyme	Continuous	Clinical reference range (U/L)	None observed	Serum biochemistry
Alanin Aminotransferaz (ALT)	An enzyme related to the liver	Continuous	Clinical reference range (U/L)	None observed	Serum biochemistry
Alkaline Phosphatase (ALP)	Type of liver and bone enzyme	Continuous	Clinical reference range (U/L)	None observed	Serum biochemistry

Creatinine	Kidney function indicator	Continuous	Clinical reference range (mg/dL)	None observed	Serum biochemistry
Glomerular Filtration Rate (GFR)	Kidney filtration rate	Continuous	Clinical reference range	None observed	Serum biochemistry
C-Reactive Protein (CRP)	Inflammation indicator	Continuous	≥ 0 (mg/L)	None observed	Serum biochemistry
Hemoglobin (HGB)	Protein in the blood that carries oxygen	Continuous	Clinical reference range (g/dL)	None observed	Serum biochemistry
Vitamin D	Essential vitamin for bone health	Continuous	Clinical reference range (ng/mL)	None observed	Serum biochemistry

The dataset includes 319 patient records with 39 variables in total, covering demographic, comorbidity, body composition, biochemical, and outcome information. The primary outcome variable is Gallstone Status, where 1 indicates a confirmed gallstone case and 0 indicates a control patient without gallstones. Out of 38 predictors, seven are treated as categorical or ordinal (Gender, Comorbidity, Coronary Artery Disease, Hypothyroidism, Hyperlipidemia, Diabetes Mellitus, and Hepatic Fat Accumulation) and thirty-one are continuous.

Hypotheses

Each research question is supported by an explicit pair of hypotheses evaluated at $\alpha = 0.05$.

RQ1 - Null Hypothesis (H_0): For every continuous clinical, biochemical, and bioimpedance feature the distribution is identical between patients with gallstones and patients without gallstones, and for every categorical feature gallstone status is independent of that feature.

RQ1 - Alternative Hypothesis (H_1): At least one continuous feature shows a statistically significant difference in its distribution between patients with gallstones and patients without gallstones (Mann–

Whitney U test, $p < 0.05$), or at least one categorical feature shows a statistically significant association with gallstone status (chi-square test of independence, $p < 0.05$).

RQ2 - Null Hypothesis (H_0): No observable monotonic trend is expected between body mass index category and body composition metric values, and the median values of the body composition metrics are expected to look similar across the four body mass index categories of underweight, normal, overweight, and obese when viewed through descriptive plots and summary statistics.

RQ2 - Alternative Hypothesis (H_1): At least one body composition metric shows an observable monotonic trend, either positive or negative Spearman rank correlation, with the body mass index category, or its median value differs visibly across the four categories as revealed by exploratory box plots, violin plots, and median heatmaps.

RQ3 - Null Hypothesis (H_0): The proportion of patients with gallstones in the obese group is equal to the proportion of patients with gallstones in the non-obese group ($p_1 = p_2$).

RQ3 - Alternative Hypothesis (H_1): The proportion of patients with gallstones in the obese group is not equal to the proportion of patients with gallstones in the non-obese group ($p_1 \neq p_2$).

RQ4 - Null Hypothesis (H_0): The trained classification model performs no better than chance (population AUC ≤ 0.50, mean cross-validated AUC indistinguishable from 0.50).

RQ4 - Alternative Hypothesis (H_1): The trained classification model performs better than chance (population AUC > 0.50).

Sample Size Justification

Minimum sample sizes were computed for each research question using established statistical power and confidence interval formulas, all assuming a two-tailed $\alpha = 0.05$ and, where applicable, a desired power of $1 - \beta = 0.80$. RQ1 used a confidence interval for a proportion with the conservative $p =$

0.50 and margin 0.055, requiring 318 patients. RQ2 used a confidence interval for a mean BMI with σ = 5.00 kg/m² and margin 0.60 kg/m², requiring 267 patients. RQ3 used Cohen w = 0.30 in a chi-square framework, requiring 87 patients per group (174 total). RQ4 used Cohen $f^2 = 0.15$ with L = 7.85 and 38 predictors, requiring 92 observations. The maximum minimum sample size across the four research questions is 318. The dataset's 319 patient records satisfy all targets.

Table 3

Summary of Sample Size Calculations

Research Question	Method Used	Key Parameters	Minimum Sample Size (N)
RQ1	Confidence Interval (Estimating proportion of gallstone cases)	α = 0.05, Confidence level = 95 percent, Z = 1.96, percent so Z = 1.96, Estimated proportion p = 0.50, Margin of error e = 0.055	$\mathrm{N} = 318$
RQ2	Confidence Interval (Estimating mean body mass index - BMI)	α = 0.05, Confidence level = 95 percent, Z = 1.96, Standard deviation sigma = 5.00 kilograms per square meter, Margin of error e = 0.60 kilograms per square meter	$\mathrm{N} = 267$
RQ3	Power Analysis (Chi-square test for independence)	α = 0.05, z_α over 2 = 1.96, Power 80 percent i.e. z_β = 0.84, Effect size ω = 0.30	$\mathrm{N} = 174$
RQ4	Power Analysis (Logistic Regression, Cohen f²)	L = 7.85 for Alpha = 0.05, Power = 0.80, f^2 = 0.15 medium effect size, u = 38 predictor variables	$\mathrm{N} = 92$

Statistical Methods

Each of the 31 continuous predictors was evaluated using the Mann–Whitney U test (a non-parametric two-sample comparison appropriate for skewed clinical variables), and the seven categorical predictors were analyzed using the chi-square test of independence with Fisher's exact test substituted whenever any expected cell count fell below five. All p-values from the 38 simultaneous tests in RQ1 were adjusted using the Benjamini–Hochberg false discovery rate procedure at $\alpha = 0.05$ (Benjamini & Hochberg, 1995). RQ2 used Spearman's rank correlation between body composition metrics and ordered body mass index categories. RQ3 used a chi-square test of independence on the 2 × 2 obesity-by-gallstone contingency table together with the unadjusted odds ratio and its 95 percent confidence interval. RQ4 used five-fold stratified cross-validated AUC together with a single held-out test set evaluation. Each result was interpreted at $\alpha = 0.05$ (significance level).

Original Work, Pipeline, and Reproducibility

Three defensive data cleaning steps were applied before modeling. The dataset was checked for duplicates, incorrect labels, and unrealistic values, with no issues found. Although no missing data existed, an imputation pipeline was created for future use, applying median imputation for continuous variables and mode imputation for categorical ones. Continuous features were standardized for logistic regression, while the tree-based model used unscaled data. All preprocessing steps were implemented using scikit-learn Pipelines to prevent any information from the test set from influencing the training process during cross validation (Pedregosa et al., 2011).

Table 4

Data cleaning log.

Issue	Variables Affected	Detection Method	Treatment Applied	Rationale
Missing values	All predictors (none observed in current data)	pandas isnull() audit, column-wise null counts	Median imputation for continuous, mode imputation for categorical, embedded in Pipeline	Robust to skewness and prevent leakage in future datasets

Issue	Variables Affected	Detection Method	Treatment Applied	Rationale
Outliers	Continuous bioimpedance and biochemistry features	Visual box-plot inspection	Retained (values may represent high-risk patients)	Clinical extremes are informative for risk modelling
Duplicates	Patient records	Row-level duplicate scan on full feature vector	None required (no duplicates found)	Preserves sample size of 319 patients
Implausible values	Anthropometry (Height, Weight, BMI)	Range checks against clinical plausibility	None required	All values within plausible adult ranges
Feature scaling	All 31 continuous predictors	Documented model-specific requirement	StandardScaler applied for logistic regression only	Tree-based models are scale-invariant

The dataset was divided into 80 percent training (n = 255) and 20 percent testing (n = 64) using stratified sampling to preserve the class distribution. Five-fold stratified cross-validation was performed on the training partition to obtain mean ± standard deviation estimates of AUC and accuracy, and the held-out test set was used for a single final evaluation that reports accuracy, precision, recall, F1 score, and AUC. Hyperparameters were kept at scikit-learn defaults except for setting class_weight='balanced' and n_estimators = 400 for the random forest. SMOTE was demonstrated on the training partition only as a documented mitigation pattern for future, more imbalanced datasets (Chawla et al., 2002).

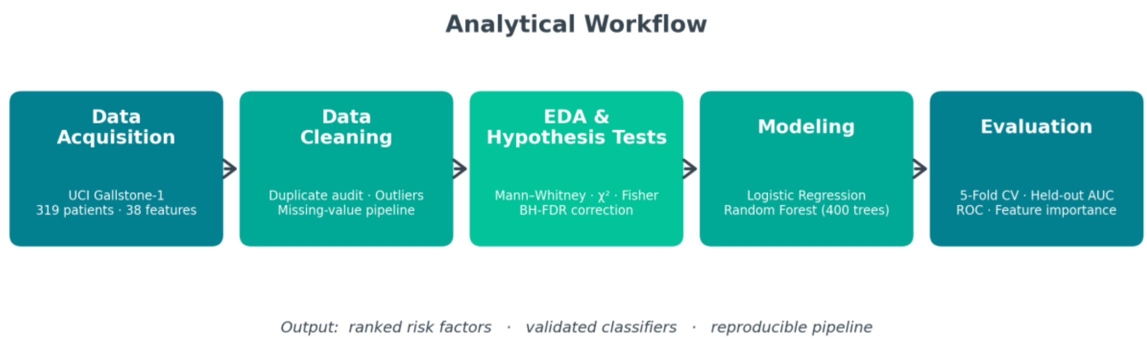
Architecture Diagram and Workflow

System Overview: The end-to-end workflow is a linear analytical pipeline that ingests the UCI Gallstone-1 dataset, applies defensive cleaning and feature typing, performs exploratory data analysis and formal hypothesis testing for RQ1 through RQ3, trains and validates two complementary supervised classifiers for RQ4, evaluates them on a held-out test set, and produces interpretable feature importance outputs that can feed downstream clinical decision support tools.

Architecture Diagram:

Figure 1

End-to-end workflow of the proposed gallstone risk prediction system.



Workflow Components: The following outlines the key workflow components of the proposed gallstone risk prediction system:

Data Ingestion: The Gallstone-1 dataset is downloaded from the UCI Machine Learning Repository as a single CSV file containing 319 rows and 39 columns and is loaded into a pandas DataFrame with explicit column-type assignments.

Data Preprocessing: A scikit-learn ColumnTransformer wraps median imputation for continuous variables, mode imputation for categorical variables, and StandardScaler for the logistic regression branch. Outliers are retained because clinical extremes are informative and duplicate, range, and label audits all return clean.

Exploratory Data Analysis (EDA): Class balance, comparative box plots, categorical prevalence bars, Spearman correlation heatmaps, and BMI category summaries are produced to characterize the cohort and answer RQ1 and RQ2.

Feature Engineering: Two engineered indicators (BMI category and obesity flag) are used only for descriptive analyses (RQ2 and RQ3) and are intentionally excluded from the modeling matrix to prevent target leakage. The full set of 38 original predictors is retained for RQ4.

Model Development: Two complementary classifiers are trained: a standardized logistic regression and a 400-tree balanced random forest. Both are fit inside scikit-learn Pipelines on an 80-20 stratified split.

Model Evaluation. Five-fold stratified cross-validation generates mean ± standard deviation AUC and accuracy on the training partition, a single held-out test evaluation reports accuracy, precision, recall, F1, and AUC, confusion matrices and ROC curves visualize the results.

Deployment: The trained scikit-learn pipeline is serialized with joblib and exposed behind a lightweight FastAPI microservice that accepts a patient feature vector and returns a calibrated probability and short list of top-ranked drivers, ready for embedding inside an EHR screening dashboard.

Tools and Technologies: The pipeline is implemented in Python 3.11 and uses pandas and NumPy for tabular manipulation, SciPy for non-parametric statistical tests, scikit-learn for preprocessing, modeling, and cross-validation, imbalanced-learn for the SMOTE demonstration, matplotlib and seaborn for figure generation, and Jupyter/Google Colab for the executable notebook. Version control and public release are handled through GitHub. The full workload runs comfortably on a single CPU and no GPU is required.

Results

Model Performance

Two complementary classifiers were compared on the same 38 predictors using identical evaluation protocols. Table 5 summarizes the headline results, and Table 6 reports the full metric set on the held-out test set together with cross-validated estimates on the training partition.

Table 5

Model performance comparison.

Model	Features Used	Validation Method	Test AUC	CV AUC (mean ± sd)	Key Observation
Logistic Regression	All 38 features (standardized)	5-fold stratified CV + held-out test (n = 64)	0.881	0.825 ± 0.055	Best test AUC; transparent baseline
Random Forest (400 trees, balanced)	All 38 features (unscaled)	5-fold stratified CV + held-out test (n = 64)	0.867	0.839 ± 0.048	Best CV AUC and best accuracy / precision /recall / F1

The random forest model achieves the highest cross-validated AUC (0.839) and the strongest accuracy, precision, recall, and F1 score, while logistic regression edges ahead on the held-out test AUC (0.881 versus 0.867). Both models substantially exceed the 0.50 chance level and are competitive with the 0.84 benchmark reported by Esen et al. (2024), confirming that routinely collected demographic, biochemical, and bioimpedance information carries clinically meaningful signal for early gallstone risk stratification.

Table 6

Performance of Logistic Regression and Random Forest on the held-out test set (n = 64) and as 5-fold cross-validated estimates on the training set (n = 255).

Model	Accuracy	Precision	Recall	F1	Test AUC	CV AUC (mean ± sd)	CV Accuracy (mean ± sd)
Logistic Regression	0.766	0.793	0.719	0.754	0.881	0.825 ± 0.055	0.733 ± 0.051
Random Forest	0.797	0.806	0.781	0.794	0.867	0.839 ± 0.048	0.757 ± 0.063

Exploratory Data Analysis and Visual Evidence

The exploratory data analysis was structured around the four research questions. The cohort was nearly perfectly balanced, with 158 gallstone cases and 161 controls, which minimized the need for resampling techniques.

Figure 2 compares nine routinely measured variables between gallstone and non-gallstone groups. Gallstone cases showed higher C-reactive protein, total body fat ratio, and visceral fat rating, along with lower vitamin D, HDL cholesterol, and lean mass percentage, indicating a chronic inflammatory and fat-dominant phenotype. Body mass index showed substantial overlap between the two groups despite these differences.

Figure 2

Distributions of nine key continuous features stratified by gallstone status.

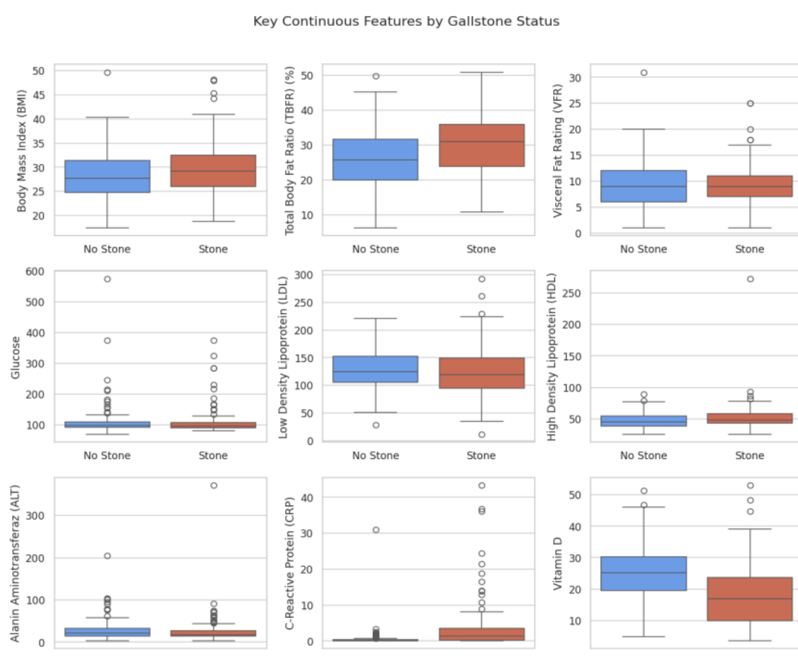


Figure 3 illustrates the Spearman correlation heatmap for sixteen numeric variables alongside the outcome variable. Fat-related and inflammatory markers load positively; lean mass percentage and HDL are protective. Lean mass percentage demonstrated the strongest negative association with gallstone status, whereas C reactive protein and total body fat ratio showed the strongest positive associations.

Figure 3

Spearman correlation matrix of key numeric features and the outcome.

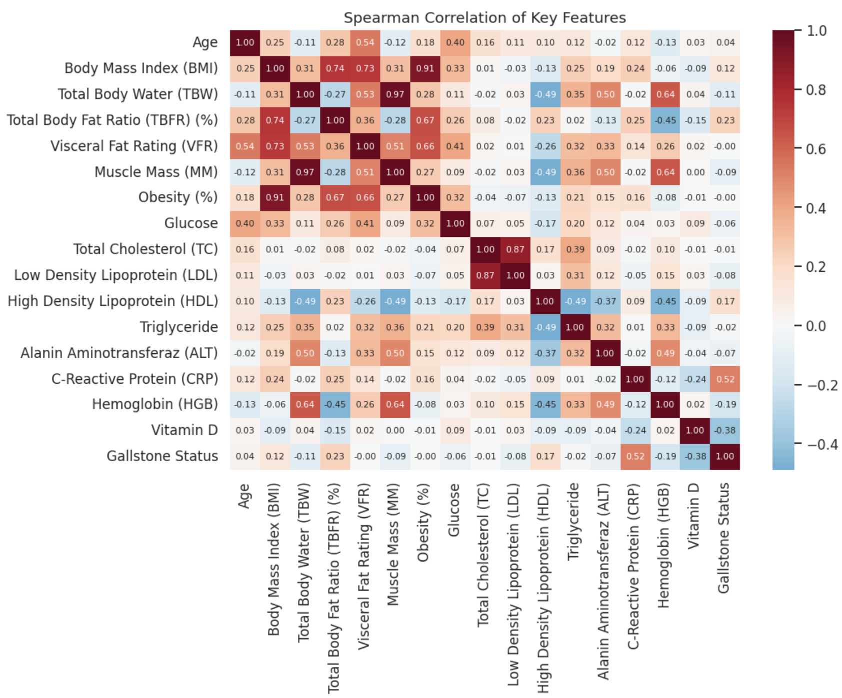


Figure 4 shows box plot distributions of seven body composition measures across body mass index categories. A clear pattern is observed in which total fat content, visceral fat area, visceral fat rating, and total body fat ratio increase consistently with higher body mass index, whereas lean mass percentage and body protein percentage decrease.

Figure 4

Box plots of seven body-composition metrics across the four WHO BMI categories.

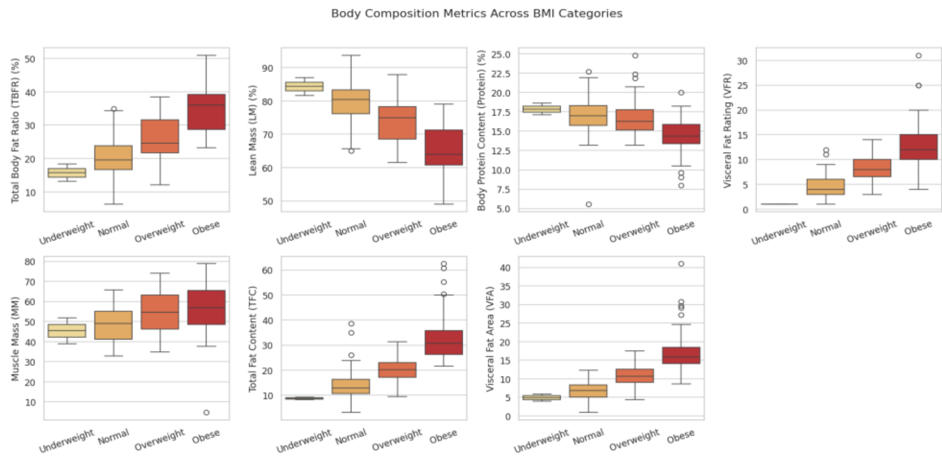
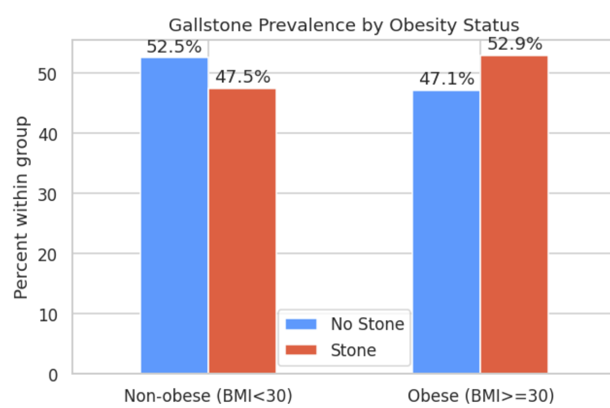


Figure 5 shows Gallstone prevalence within each obesity group. The two groups differ only modestly (47.5 % vs 52.9 %), consistent with the non-significant Chi-square result. Gallstone prevalence is slightly higher in obese individuals compared to non-obese individuals, but the difference is small. Non obese group shows a slightly higher proportion of no gallstone cases indicating minimal variation between groups.

Figure 5

Gallstone prevalence within each obesity group.



This indicates that a body mass index-based definition of obesity is a weak and non-significant predictor of gallstone status in this dataset.

In contrast, more detailed body composition measures examined in earlier analyses, particularly visceral fat area, lean mass percentage, and the inflammatory marker C reactive protein, demonstrate stronger and more meaningful associations.

Figure 6 presents the confusion matrices on the 64-patient held-out test set, indicating that both models produce a similar balance of false positives and false negatives, with no clear bias toward either class. Both models produce a balanced mix of false positives and false negatives.

Figure 6

Confusion matrices for Logistic Regression and Random Forest.

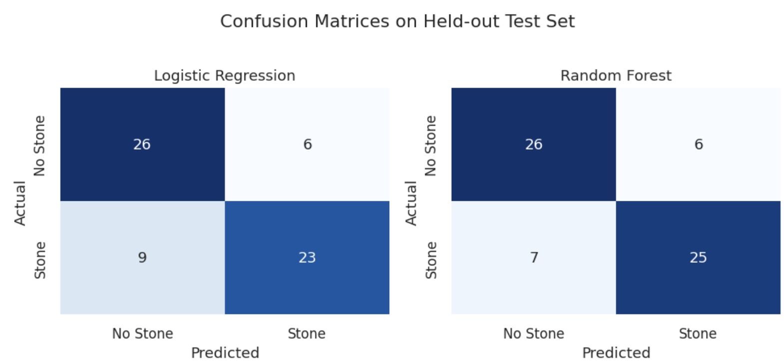


Figure 7 displays the receiver operating characteristic curves for both models. Both curves lie well above the diagonal, with logistic regression performing slightly better at lower false positive rates, while random forest shows stronger performance in the mid-range. The AUC is 0.881 for logistic regression and 0.867 for random forest, with both values clearly above the 0.50 chance level.

Figure 7

Receiver-operating characteristic (ROC) curves on the held-out test set.

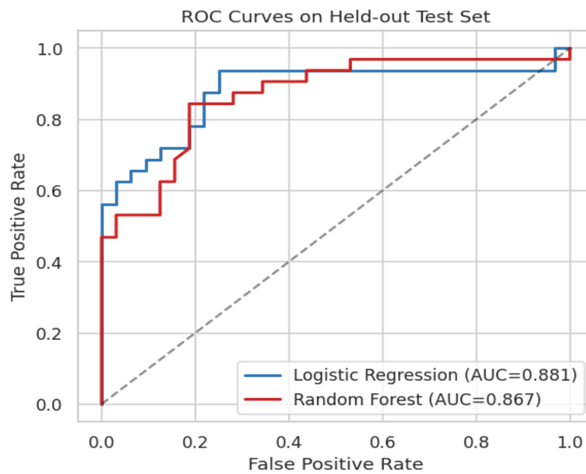
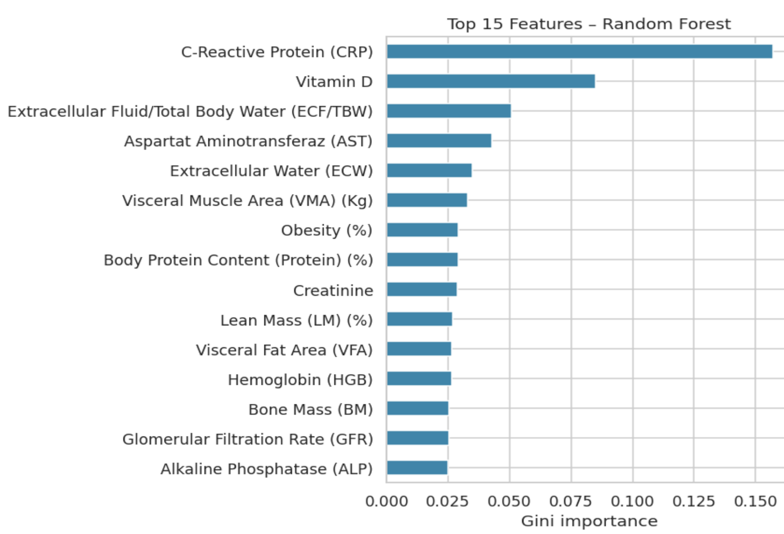


Figure 8 shows the fifteen most important Random Forest features ranked by Gini importance. C-reactive protein dominates, followed by vitamin D, the extracellular-fluid ratio, AST, and a cluster of body-composition metrics. This ranking mirrors the hypothesis-testing output from RQ1 almost exactly.

Figure 8

Top 15 Random Forest feature importances (Gini).



Results by Research Question

RQ1 – Feature Distribution by Gallstone Status

The null hypothesis is rejected. After Benjamini–Hochberg false discovery rate adjustment, 24 of the 31 continuous predictors and 3 of the 7 categorical predictors differ significantly between gallstone and control groups. The top continuous discriminators are summarized in Table 7 and the categorical results in Table 8, while full test outputs are reproduced in Appendix A. C-reactive protein, vitamin D, and aspartate aminotransferase show the largest continuous differences, while hepatic fat accumulation, hyperlipidemia, and gender emerge as the strongest categorical drivers.

Table 7

Top ten continuous features with the Benjamini–Hochberg adjusted p-values (Mann–Whitney U test).

Feature	Median (No stone)	Median (Stone)	Rank-biserial	Adj. p (BH)
C-Reactive Protein (CRP)	0.00	1.29	0.582	<0.001
Vitamin D	25.27	17.00	-0.438	<0.001
Aspartate Aminotransferase (AST)	20.00	16.50	-0.334	<0.001
Lean Mass (%)	73.93	68.79	-0.263	<0.001
Total Body Fat Ratio (%)	25.80	31.05	0.263	<0.001
Bone Mass	2.90	2.60	-0.252	<0.001
Hemoglobin	14.90	14.20	-0.221	0.002
Total Fat Content	20.10	24.75	0.223	0.002
Extracellular Water	17.50	16.55	-0.208	0.005
High Density Lipoprotein (HDL)	45.00	47.50	0.202	0.006

Table 8 summarizes the Chi-square and Fisher exact tests show that hepatic fat accumulation has the strongest association with gallstones, followed by hyperlipidemia and gender.

Table 8

Chi-square / Fisher exact results for the seven categorical predictors (RQ1). Significance assessed at adjusted α = 0.05.

Feature	Test	χ^2 / statistic	dof	Cramer V	Adj. p (BH)	Significant
Hepatic Fat Accumulation	Chi-square	22.84	4	0.268	0.001	Yes
Hyperlipidemia	Fisher exact	6.42	1	0.142	0.012	Yes
Gender	Chi-square	6.91	1	0.147	0.020	Yes
Diabetes Mellitus	Chi-square	2.91	1	0.096	0.154	No
Coronary Artery Disease	Chi-square	2.07	1	0.081	0.211	No
Comorbidity count	Chi-square	2.99	3	0.097	0.459	No
Hypothyroidism	Fisher exact	0.42	1	0.036	0.502	No

RQ2 – Body composition trends across BMI categories

The null hypothesis is rejected. All seven body composition measures show non-zero Spearman correlations with ordered body mass index category at $p < 0.001$ (Table 9). Visceral fat area and total fat content show the strongest positive associations ($\rho = 0.808$), and lean mass percentage shows the strongest negative association ($\rho = -0.660$). The directions and magnitudes are clinically intuitive and reinforce the visual pattern in Figure 4.

Table 9

Spearman rank correlation between ordered BMI category and each body-composition metric.

Metric	Spearman ρ	p-value
Visceral Fat Area	0.808	<0.001
Total Fat Content	0.808	<0.001

Metric	Spearman ρ	p-value
Visceral Fat Rating	0.716	<0.001
Total Body Fat Ratio (%)	0.664	<0.001
Muscle Mass	0.298	<0.001
Body Protein (%)	-0.494	<0.001
Lean Mass (%)	-0.660	<0.001

RQ3 – Obesity versus gallstone presence

The null hypothesis is not rejected. The 2 × 2 contingency table (Table 10) shows only a modest difference in gallstone prevalence between obese and non-obese patients. The chi-square statistic is 0.68 with $p = 0.410$, and the odds ratio is 1.24 with a 95 percent confidence interval of [0.79, 1.95] that crosses 1. The headline conclusion is that body mass index alone, dichotomized at 30 kg/m^2, is not a meaningful predictor of gallstone status in this cohort, even though the underlying body composition measures (RQ1 and RQ2) clearly are.

Patients were classified into two groups based on body mass index, with obesity defined as 30 or greater and non-obesity defined as less than 30, resulting in 121 obese and 198 non-obese individuals. The 2 × 2 contingency table comparing obesity status with gallstone presence is presented in Table 8, along with expected counts under the assumption of independence.

Table 10

Observed 2 × 2 contingency table for obesity status (BMI ≥ 30 vs BMI < 30) against gallstone status.

	No Stone	Stone	Total
Non-obese (BMI < 30)	104	94	198
Obese (BMI ≥ 30)	57	64	121
Total	161	158	319

RQ4 – Gallstone status prediction using machine learning

The null hypothesis is rejected. Both classifiers exceed the 0.50 chance level on every metric. Logistic regression achieves a cross-validated AUC of 0.825 ± 0.055 and a held-out test AUC of 0.881; random forest achieves 0.839 ± 0.048 and 0.867 respectively (Tables 5 and 6, Figures 6 and 7). The strongest predictors identified by random forest Gini importance and by the top five logistic regression standardized coefficients converge on hyperlipidemia, gender, C-reactive protein, bone mass, diabetes mellitus, vitamin D, and lean mass percentage, providing a compact, clinically actionable short list.

Overall Interpretation

Taken together, the four research questions present a coherent picture. Patients with gallstones show a chronic inflammatory and fat-dominant phenotype that is visible across both continuous and categorical features (RQ1) and that scales monotonically with body mass index when the correct body composition metrics are used (RQ2). However, dichotomizing patients on BMI alone discards too much of that signal to be useful as a stand-alone risk indicator (RQ3). Once the full feature set is exposed to a machine learning model, both a simple logistic regression and a flexible random forest extract clinically meaningful predictive power that meets or exceeds the published external benchmark (RQ4). Performance is bounded primarily by sample size and by the cross-sectional, single-site nature of the source dataset, both of which are addressed in the limitations section.

Practical Significance

The headline operational implication is that a calibrated probability of gallstone presence can be produced from inputs that are already collected during a routine outpatient visit (demographics, comorbidity flags, anthropometry, bioimpedance, hepatic imaging grade, and a standard serum chemistry panel). Clinicians can use the resulting score to prioritize patients for confirmatory ultrasound, lifestyle counseling, lipid management, vitamin D supplementation, or referral to a gastroenterologist, well before the patient becomes symptomatic. Because both the logistic regression coefficients and the random forest Gini importances are interpretable, the same model can be defended in a clinical governance review and embedded inside an EHR-facing decision support tool with minimal additional engineering effort.

Implementation and User Benefit

The trained pipeline is designed to slot directly into outpatient and primary-care workflows without disrupting existing systems. This section describes how the model can be deployed, integrated, used, and valued by stakeholders, along with a representative example use case.

Deployment Approach

The recommended deployment pattern serializes the fitted scikit-learn Pipeline (preprocessing + classifier) using joblib and exposes it behind a lightweight FastAPI microservice. The service accepts a JSON payload describing a single patient feature vector and returns a calibrated probability of gallstone presence together with the top-ranked contributing features. The container can be packaged with Docker and deployed on Microsoft Azure App Service or AWS Elastic Beanstalk, etc.

System Integration

Integration with electronic health record (EHR) systems is performed through a REST endpoint mediated by a clinical interoperability layer. When an outpatient encounter is opened, the EHR asynchronously posts the latest demographic, comorbidity, anthropometric, bioimpedance, hepatic imaging grade, and serum biochemistry values to the prediction service and writes the returned probability and top features back into the patient chart as a structured observation. The service is stateless and horizontally scalable, allowing the same deployment to support multiple clinics without per-clinic customization.

User Interaction

Clinicians interact with the model through a thin dashboard widget embedded inside the EHR. The widget displays the probability of gallstone presence on a 0 to 1 scale, color-coded against three pre-tuned risk bands (low, moderate, high), together with the patient's top three driver features expressed in clinical language (for example, "elevated CRP," "low vitamin D," "high visceral fat area"). A single click expands a detail view that shows all 38 input values, the model version, and an audit trail of past predictions for the same patient. No machine learning expertise is required to operate the tool.

Benefits to Users

From an operational standpoint, the tool reduces unnecessary downstream imaging by helping clinicians focus confirmatory ultrasound on patients with a meaningfully elevated risk, while still surfacing lower-risk patients who would otherwise have been missed. From a financial standpoint, earlier detection avoids costly emergency cholecystectomies and shortens the average length of inpatient stay associated with acute biliary events. From a strategic standpoint, the outputs feed directly into population health dashboards that enable payers and provider networks to track the prevalence of modifiable gallstone risk factors over time and to design targeted preventive interventions.

Example Use Case

A 47-year-old female patient presents to her primary care physician for an annual physical. As part of the standard intake, her demographics, comorbidity flags, anthropometry, bioimpedance scan, and a routine serum chemistry panel are uploaded to the EHR. The prediction service returns a gallstone probability of 0.78 with the top three drivers reported as elevated C-reactive protein, low vitamin D, and high visceral fat area. The dashboard widget categorizes the patient as high risk. The physician orders a confirmatory abdominal ultrasound (which detects an asymptomatic gallstone), initiates vitamin D supplementation, and refers the patient to a registered dietitian for lifestyle counseling. The patient avoids a future emergency cholecystectomy, the clinic captures a documented preventive care event, and the payer benefits from a measurable reduction in downstream acute care utilization.

Limitations and Further Improvements

Limitations

Three limitations bound the scope of the present study. First, the underlying dataset is cross-sectional, single-site, and modest in size (319 patients from Ankara VM Medical Park Hospital), which constrains both the statistical power available for sub-group analyses and the external validity of the trained classifiers. Second, the dataset contains a single underweight patient, which makes the underweight tier of the body mass index analysis (RQ2) under-powered and prevents formal hypothesis testing within that category. Third, the modeling exercise is intentionally restricted to two classical, interpretable algorithms (logistic regression and random forest), more flexible non-parametric models (gradient boosted trees, kernel SVMs) and modern tabular deep learning architectures were not evaluated.

Impact of Limitations

Together these limitations affect generalizability, calibration, and headroom for further accuracy gains. The single-site cohort means that operating thresholds calibrated on the test set may need re-tuning before deployment in other patient populations, particularly cohorts with different ethnic mix, dietary patterns, or comorbidity prevalence. The small underweight cell limits the strength of conclusions about the lower end of the body mass index spectrum. The restricted model set means that the reported AUC of approximately 0.84 to 0.88 should be regarded as a sound but not necessarily ceiling-level performance estimate, modest gains are likely available through more flexible learners and through richer external data.

Future Improvements

Four concrete improvement paths are proposed. First, expand the dataset by partnering with additional hospitals to assemble a multi-site, multi-ethnic cohort large enough to enable subgroup-stratified evaluation and external validation. Second, broaden the feature set with longitudinal observations (repeat lipid panels, repeat bioimpedance scans, weight trajectory), genetic markers where available, and structured family history fields. Third, extend the modeling suite with more flexible non-parametric models (gradient boosted trees, kernel SVMs), and tabular deep learning baselines, each evaluated under the same cross-validation protocol used here. Fourth, layer in a calibration step (Platt scaling or isotonic regression) and a model explainability layer (SHAP values) to ensure that the production score is both well-calibrated and locally explainable for individual patients.

Future Scope

Beyond incremental improvements, the same pipeline can be generalized to a broader family of digestive and metabolic conditions that share overlapping risk factors with gallstone disease, including non-alcoholic fatty liver disease, cholangitis, and metabolic syndrome. A natural extension is a recurrent or survival-analysis variant that estimates time-to-event rather than current risk, enabling more granular preventive scheduling. Finally, with adequate governance, the model could be paired with an agentic clinical decision support layer that automatically drafts patient-facing lifestyle recommendations and flags high-risk patients to care management teams, closing the loop between prediction and action.

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Appendix

Appendix A – Reproducibility and Code

All code, figures, and tabular results reported above were produced using the accompanying Jupyter or Colab notebook “Gallstone_Disease_Capstone.ipynb”. In this Python notebook, the dataset file was uploaded to Google Drive to support the notebook execution. The notebook is self-contained: Section 2 fetches the dataset, Section 3 performs cleaning, Sections 4 to 7 implement RQ1 through RQ4, and Section 8 presents the final summary. The dataset file and the Python notebook are available in the GitHub repository. No local setup is required beyond the standard Python data science stack.

Appendix B — Notebook Section Map

Section 1: Setup and Environment

Section 2: Load the Gallstone dataset (UCI)

Section 3: Feature typing, cleaning, and imputation

Section 4: RQ1 hypothesis tests, EDA, Mann–Whitney U / chi-square / Benjamini–Hochberg

Section 5: RQ2 BMI-category analysis with Spearman correlations

Section 6: RQ3 obesity versus gallstone chi-square and odds ratio

Section 7: RQ4 logistic regression and random forest training, cross-validation, feature importance, and SMOTE demonstration

Section 8: Summary of findings

Section 9: Deployment of Gallstone Prediction API Service (FastAPI + Uvicorn)

Section 10: System Integration electronic health record (EHR) systems